

1.		5
1.1.		5
1.2.		5
2.		6
2.1.		6
2.2.		6
2.3.		6
2.4.		7
2.5.		7
2.6.		7
2.7.		7
2.8.		8
3.		9
3.1.		9
3.2.		9
3.3.		9
3.4.		10
3.5.		10
3.6.		12
3.7.		12
3.8.		12
3.9.		13
3.10.		14
3.11.		14
3.12.		14
3.13.		15

3.14.		15
3.15.		15
3.16.		16
3.17.		16
3.18.		17
3.19.		18
3.20.		18
3.21.		19
3.22.		20
3.23. n		20
4.		21
4.1.		21
4.2.		21
4.3.		22
4.4.		23
4.5.		24
4.6.		25
4.7.		25
4.8.		26
4.9.		27
4.10.		28
4.11.		28
5.		30
5.1.		30
5.2.		30
5.3.		30
5.4.		31
5.5.		32
5.6.		32
5.7.		33
5.8.		34

6.		36
6.1.		36
6.2.		36
6.3.		37
6.4.		37
6.5.		38
6.6.		38
6.7.		39
6.8.		40
6.9.		41
6.10.		41
6.11.		42
6.12.		42
6.13.		43
6.14.		43
6.15.		43
7.		44
7.1.		44
7.2.		44
7.3.		45
7.4.		45
7.5.		45
7.6.		45
7.7.		46
7.8.		46
7.9.		46
7.10.		46
7.11.		47
7.12.		47
7.13.	2	47
7.14.		48

7.15. 48

1.

1.1.

2022

pp. 40 - 41

1.2.

2022

pp. 43, 47 - 54

12

$X\text{cm}^2$

$Y\text{ g}$

(1)

$Y = aX + b$

(2)

(3)

$(X, Y) = (11, 23), \quad (23, 49), \quad (5, 10), \quad (50, 115), \quad (32, 64), \quad (21, 45)$
 $(16, 35), \quad (39, 84), \quad (25, 50), \quad (45, 100), \quad (5, 11), \quad (34, 70)$

2.4.

2023

pp. 81, 86

$$f(x) = \begin{cases} \frac{c}{x^3} & (x \geq 10) \\ 0 & (x < 10) \end{cases}$$

(1)

c

(2)

$F(x)$

(3)

$P(x \geq a) = \frac{1}{2}$

a

2.5.

2024

pp. 81, 86

c

$$f(x) = \begin{cases} c(1-x) & (0 \leq x \leq 1) \\ 0 & (\text{otherwise}) \end{cases}$$

(1)

c

(2)

$P\left(\frac{1}{2} < x \leq \frac{2}{3}\right)$

(3)

$F(x)$

(4)

$P\left(\frac{1}{2} \leq x < 3\right)$

2.6.

2022

2023

2024

pp. 134, 151, 180

(1)

(2)

(3)

(4)

2.7.

2023

2023

pp. 134, 151, 180

$X_1, X_2, \dots, X_n$

Y

2

“

” “ ”

(1)

$X_1, X_2, \dots, X_n$

Y

(2)

2

/

(3)

- (4)
- (5)
- (6)1

2.8.

2025						
(1)	X	μ	σ^2			
(2)	200		60	6		42
	78					

3.

3.1.

2024

p. 107

6

4

1

2

(1) 2

(2) 2

(1) $\frac{6}{10} \cdot \frac{5}{9} = \frac{1}{3}$

(2) $\frac{6}{10} \cdot \frac{5}{9} + \frac{4}{10} \cdot \frac{6}{9} = \frac{3}{5}$

3.2.

2026

p. 107

3

5

1	2	3	4	2
X_1	X_2	$Y = X_1 - X_2 $		
(1) Y				
(2) Y				
(3) Y				

(1)	Y	0	1	2	3	
	P	$1/4$	$3/8$	$1/4$	$1/8$	1

$$(2) E[Y] = \frac{1}{4} \cdot 0 + \frac{3}{8} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{8} \cdot 3 = \frac{5}{4}$$

$$(3) \text{Var}[Y] = E[Y^2] - (E[Y])^2 = \left(\frac{1}{4} \cdot 0^2 + \frac{3}{8} \cdot 1^2 + \frac{1}{4} \cdot 2^2 + \frac{1}{8} \cdot 3^2 \right) - \left(\frac{5}{4} \right)^2 = \frac{5}{2} - \frac{25}{16} = \frac{15}{16}$$

3.4.

2023

2025

pp. 113 - 114

 (X, Y)

$$f(x, y) = \begin{cases} 6e^{-2x-3y} & (x \geq 0, y \geq 0) \\ 0 & (\text{otherwise}) \end{cases}$$

 $X \quad Y$

$$x \geq 0 \quad y \geq 0$$

$$f_1(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_0^{\infty} 6e^{-2x-3y} dy = 6e^{-2x} \left[-\frac{1}{3}e^{-3y} \right]_0^{\infty} = 2e^{-2x}$$

$$f_2(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^{\infty} 6e^{-2x-3y} dx = 6e^{-3y} \left[-\frac{1}{2}e^{-2x} \right]_0^{\infty} = 3e^{-3y}$$

$$f_1(x) = \begin{cases} 2e^{-2x} & (x \geq 0) \\ 0 & (x < 0) \end{cases} \quad f_2(y) = \begin{cases} 3e^{-3y} & (y \geq 0) \\ 0 & (y < 0) \end{cases}$$

$$x \geq 0 \quad y \geq 0$$

$$f(x, y) = f_1(x)f_2(y)$$

 $X \quad Y$

3.5.

2026

$X \quad Y$

$$\int_0^\infty e^{-x} dx = 1, \quad \int_0^\infty x e^{-x} dx = 1$$

$$(1) \quad f(x, y) = \begin{cases} x e^{-x-y} & (x \geq 0, y \geq 0) \\ 0 & (\text{otherwise}) \end{cases}$$

$$(a) \quad f_X(x) \quad f_Y(y)$$

$$(b) \quad X \quad Y$$

$$(2) \quad f(x, y) = \begin{cases} c(x+y)e^{-x-y} & (x \geq 0, y \geq 0) \\ 0 & (\text{otherwise}) \end{cases}$$

$$(a) \quad c$$

$$(b) \quad X \quad Y$$

$$(1) \quad (a) \quad x \geq 0 \quad y \geq 0$$

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy = x e^{-x} \underbrace{\int_0^{\infty} e^{-y} dy}_{=1} = x e^{-x}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx = e^{-y} \underbrace{\int_0^{\infty} x e^{-x} dx}_{=1} = e^{-y}$$

$$f_X(x) = \begin{cases} x e^{-x} & (x \geq 0) \\ 0 & (\text{otherwise}) \end{cases}, \quad f_Y(y) = \begin{cases} e^{-y} & (y \geq 0) \\ 0 & (\text{otherwise}) \end{cases}$$

$$(b) \quad f(x, y) = x e^{-x} e^{-y} = f_X(x) f_Y(y) \quad X \quad Y$$

$$(2) \quad (a) \quad \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

$$\int_0^{\infty} \int_0^{\infty} c(x+y) e^{-x-y} dx dy = c \underbrace{\int_0^{\infty} x e^{-x} dx}_{=1} \underbrace{\int_0^{\infty} e^{-y} dy}_{=1} + c \underbrace{\int_0^{\infty} e^{-x} dx}_{=1} \underbrace{\int_0^{\infty} y e^{-y} dy}_{=1} = 2c$$

$$c = \frac{1}{2}$$

$$(b) \quad x \geq 0, y \geq 0$$

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy = \frac{1}{2} x e^{-x} \underbrace{\int_0^{\infty} e^{-y} dy}_{=1} + \frac{1}{2} e^{-x} \underbrace{\int_0^{\infty} y e^{-y} dy}_{=1} = \frac{1}{2} (x+1) e^{-x}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)dx = \frac{1}{2}e^{-y} \underbrace{\int_0^{\infty} xe^{-x}dx}_{=1} + \frac{1}{2}ye^{-y} \underbrace{\int_0^{\infty} e^{-x}dx}_{=1} = \frac{1}{2}(y+1)e^{-y}$$

$x = y = 0$

$f(x,y) \neq f_X(x)f_Y(y)$

$X \quad Y$

3.6.

2022

pp. 103 - 104, 113

1	4	2	2	1
	$X = 0$	$X = 1$	2	
$Y = 0$		$Y = 1$		
(1)	(X,Y)			
(2)	X	Y		

(1)

$X \backslash Y$	0	1	
0	$\frac{2}{5}$	$\frac{2}{15}$	$\frac{2}{3}$
1	$\frac{4}{15}$	$\frac{1}{15}$	$\frac{1}{3}$
	$\frac{2}{3}$	$\frac{1}{3}$	1

(2)

$P(X = 0, Y = 0) \neq P(X = 0)P(Y = 0)$

3.7.

2022

pp. 67, 114 - 115

2	A	B
---	-----	-----

A

B

$P(A \cap B) = P(A)P(B)$

A

$P(A \cap B) = 0$

2

2

3.8.

2022

2023

2026

p. 115

3

$\frac{1}{2}$

(1)

1

2

(2)

1

2

A

2

3

B

A

B

(3)

$P(A|B)$

(1)

$\Omega = (M, F)^3$

(M, M, F)

1

2

3

$\equiv \Omega$

"M

"

$\equiv$

$\Omega_0 = [(F, F, F)]_{\equiv}$

$\Omega_1 = [(M, F, F)]_{\equiv}$

$\Omega_2 = [(M, M, F)]_{\equiv}$

$\Omega_3 = [(M, M, M)]_{\equiv}$

$|\Omega_0| = \binom{3}{0} =$

1

$|\Omega_1| = \binom{3}{1} = 3$

$|\Omega_2| = \binom{3}{2} = 3$

$|\Omega_3| = \binom{3}{3} = 1$

$|\Omega| = 8$

$\frac{4/8}{7/8} = \frac{4}{7}$

(2)

A

$A = \{(M, M, F), (M, M, M)\}$

B

$B = \{(F, M, M), (M, M, M)\}$

$A \cap B$

$A \cap B = \{(M, M, M)\}$

$|\Omega|$

$P(A) = \frac{1}{4}$

$P(B) =$

$\frac{1}{4}$

$P(A \cap B) = \frac{1}{8}$

A

B

$P(A \cap B) =$

$P(A)P(B)$

A

B

(3)

$P(A|$

$$\begin{aligned}
 & \bullet P(B) = P(X + Y = 7) = \frac{1}{6} \\
 & \bullet P(A \cap B) = P(X = 4, X + Y = 7) = P(X = 4, Y = 3) = \frac{1}{36} \\
 & P(A \cap B) = P(A)P(B) \quad A \quad B \\
 (2) \quad & \bullet P(A) = \frac{1}{6} \\
 & \bullet P(C) = P(X + Y = 9) = \frac{1}{9} \\
 & \bullet P(A \cap C) = P(X = 4, X + Y = 9) = P(X = 4, Y = 5) = \frac{1}{36} \\
 & P(A \cap C) \neq P(A)P(C) \quad A \quad C
 \end{aligned}$$

3.10.

2023

pp. 72 - 73, 109 - 110, 114 - 115

$$\begin{aligned}
 & 2 \quad E \quad F \quad P(E) = a \quad P(F) = b \\
 (1) \quad & P(E \cup F) \quad a \quad b \\
 (2) \quad & P(E \cap F^c | E \cup F) \quad a \quad b \quad F^c \quad F
 \end{aligned}$$

$$\begin{aligned}
 (1) \quad & P(E \cup F) = a + b - ab \\
 (2) \quad & P(E \cap F^c | E \cup F) = \frac{P(E \cap F^c)}{P(E \cup F)} = \frac{a - ab}{a + b - ab}
 \end{aligned}$$

3.11.

2025

1 \quad 5

$$X \quad Y \quad Z = X - Y$$

$$\begin{aligned}
 (1) \quad & X \quad Y \\
 (2) \quad & Z \\
 (3) \quad & Z \\
 (4) \quad & Z
 \end{aligned}$$

3.12.

2025

pp. 118 - 119

$$(1) \quad 2 \quad A \quad B$$

(2)	500	1	X	X
2	K_1	K_2	$3 : 1$	
		K_1, K_2	0.98 0.90	
			0.01	
			K_1	

3.13.

2022	2023	pp. 118 - 119	
	3	5	1
	2		
(1)	1		
(2)	2	1	

3.14.

2023	pp. 118 - 119	
	1	10%
		3
(1)		
(2)	2	

(1)

$$\frac{9}{10} \cdot \frac{9}{10} \cdot \frac{9}{10} = 0.7290$$

(2)

$$\frac{\frac{9}{10} \cdot \frac{1}{10}}{\frac{1}{10} + \frac{9}{10} \cdot \frac{1}{10} + \frac{9}{10} \cdot \frac{9}{10} \cdot \frac{1}{10}} = \frac{90}{271} \approx 0.3321$$

3.15.

2022	pp. 118 - 119			
	5	4	3	
1		100 g	45 g	30 g
(1)	1			
	40 g			

(2)

2

100 g

2

(3)

3

150 g

3

(1)

$$\frac{\binom{5}{1}}{\binom{12}{1} - \binom{3}{1}} = \frac{5}{9}$$

(2)

$$\frac{\binom{5}{2}}{\binom{12}{2} - \binom{7}{2}} = \frac{10}{66 - 21} = \frac{2}{9}$$

(3)

$$\frac{\binom{5}{3}}{\binom{12}{3} - \binom{7}{3}} = \frac{10}{220 - 35} = \frac{10}{185} = \frac{2}{37}$$

3.16.

2022

2024

pp. 118 - 119

A

0.1%

A

/

99%

10%

(1)

(2)

A

1

A

A

B

2

C

(1)

$$P(A|B) = \frac{P(A)P(B|A)}{P(A)P(B|A) + P(A^c)P(B|A^c)} = \frac{0.001 \cdot 0.99}{0.001 \cdot 0.99 + 0.999 \cdot 0.10} = \frac{99}{10089} \approx 0.0098$$

(2)

$$P(A|C) = \frac{P(A)P(C|A)}{P(A)P(C|A) + P(A^c)P(C|A^c)} = \frac{0.001 \cdot 0.99^2}{0.001 \cdot 0.99^2 + 0.999 \cdot 0.10^2} = \frac{9801}{109701} \approx 0.0893$$

3.17.

2023

pp. 118 - 119

99%

0.5%

99%

1

(4)

	A	B	
(1) $P(A \cap B) = P(A)P(B A) = 0.005 \cdot 0.99 = 0.00495$			
(2) $P(A^c \cap B) = P(A^c)P(B A^c) = 0.995 \cdot 0.01 = 0.00995$			
(3) $P(B) = P(A \cap B) + P(A^c \cap B) = 0.00495 + 0.00995 = 0.0149$			
			$P(A^c B) = \frac{P(A^c \cap B)}{P(B)} = \frac{0.00995}{0.0149} \approx 0.6678$
(4) $P(A B^c) = \frac{P(A \cap B^c)}{P(B^c)} = \frac{P(A)P(B^c A)}{1 - P(B)} = \frac{0.005 \cdot 0.01}{0.9851} \approx 0.0001$			

2023 pp. 118 - 119

S
 K
 • S K 99% 1%
 • S K 99% 1%

(2) (1)

$$\begin{array}{ccccc}
 (1) & S & & A & K & & B \\
 & & & & & & \\
 P(A|B) = & \frac{P(A)P(B|A)}{P(A)P(B|A) + P(A^c)P(B|A^c)} = & \frac{\frac{1}{10000} \cdot \frac{99}{100}}{\frac{1}{10000} \cdot \frac{99}{100} + \frac{9999}{10000} \cdot \frac{1}{100}} = & \frac{99}{10098} \approx 0.0098
 \end{array}$$

3.19.

2025

pp. 118 - 119

A

B

(1) $P(A|B) = 0.98 \quad P(A|B^c) = 0.005 \quad P(B) = 0.001$

(2) $P(A|B^c) = 0.003 \quad P(B) = 0.0005 \qquad P(B|A) \geq 0.142 \qquad P(A|B)$

(1)

$$P(B|A) = \frac{P(B)P(A|B)}{P(B)P(A|B) + P(B^c)P(A|B^c)} = \frac{0.001 \cdot 0.98}{0.001 \cdot 0.98 + 0.999 \cdot 0.005} = \frac{980}{5975} \approx 0.1640$$

(2) (1)

$$P(B|A) = \frac{0.0005 \cdot P(A|B)}{0.0005 \cdot P(A|B) + 0.9995 \cdot 0.003} \geq 0.142$$
$$P(A|B) \geq \frac{1}{0.0005} \cdot \frac{1}{1 - 0.142} + 0.142 \cdot 0.9995 \cdot 0.003 \approx 0.9925$$

3.20.

pp. 118 - 119

500

K_1

1

K_2

XXX

$K_1 : K_2 = 3 : 1$

2

K_1

0.98

K_2

0.90

0.01

(1)

 K_1

(2) (1)

$$\begin{array}{c}
 K_1 \quad K_2 \qquad \qquad \qquad A_1 \quad A_2 \\
 A_0 \qquad \qquad \qquad B \\
 (1) \quad P(A_1) = \frac{1}{500} \cdot \frac{3}{4} = \frac{3}{2000} \quad P(A_2) = \frac{1}{500} \cdot \frac{1}{4} = \frac{1}{2000} \quad P(A_0) = \frac{499}{500} = \frac{1996}{2000} \\
 P(B) = \frac{\frac{3}{2000} \cdot \frac{98}{100}}{P(A_1 \cap B)} + \frac{\frac{1}{2000} \cdot \frac{90}{100}}{P(A_2 \cap B)} + \frac{\frac{1996}{2000} \cdot \frac{1}{100}}{P(A_0 \cap B)} = \frac{294 + 90 + 1996}{200000} = \frac{2380}{200000} \\
 P(A_1|B) = \frac{P(A_1 \cap B)}{P(B)} = \frac{294}{2380} \approx 0.1235 \\
 (2)
 \end{array}$$

3.21.

2023

pp. 122 - 124

$$p \quad 0 < p < 1$$

$$X \quad Y$$

$$f(k) = \begin{cases} p & (k=1) \\ 1-p & (k=-1) \end{cases}$$

$$Z = XY$$

$$(1) \quad E[Z]$$

$$(2) \quad X \quad Z \quad E[(X - E[X])(Z - E[Z])]$$

$$(3) \quad X \quad Z \quad p \quad p \quad Y \quad Z$$

$$(4) \quad (3) \quad p \quad \Pr[X + Y + Z \leq 2]$$

$$(1) \quad E[X] = E[Y] = p - (1-p) = 2p - 1 \quad E[Z] = E[XY] = E[X]E[Y] = (2p - 1)^2$$

$$(2) \quad E[XZ] = E[X^2Y] = E[Y]$$

$$\text{Cov}(X, Z) = E[XZ] - E[X]E[Z] = (2p - 1)[1 - (2p - 1)^2] = 4p(1 - p)(2p - 1)$$

$$(3) \quad \text{Cov}(X, Z) = 0 \quad \text{---} \quad 0 < p < 1 \quad p = \frac{1}{2}$$

$$P(X = a, Y = b) = P(X = a)P(Y = b) = \frac{1}{4}$$

$$(4) \quad (X, Y) = (1, 1) \quad X + Y + Z = 3 > 2$$

$$\Pr[X + Y + Z \leq 2] = 1 - \Pr[X + Y + Z > 2] = \frac{3}{4}$$

2023

2024

pp. 125 - 126

2

$$X \quad Y$$

$$(1) \quad a$$

(2)	X	Y	$\text{Cov}(X, Y)$
-----	-----	-----	--------------------

$$(3) \quad X \quad Y$$

	$Y = 1$	$Y = 10$	$Y = 100$
$X = 1$	a	a	a
$X = 2$	0	a	a
$X = 3$	a	a	a

$$(1) \qquad 8a = 1 \qquad a = \frac{1}{8}$$

(2) $X \rightarrow Y$

- $E[X] = \frac{3}{8} \cdot 1 + \frac{2}{8} \cdot 2 + \frac{3}{8} \cdot 3 = 2$

- $E[Y] = \frac{2}{8} \cdot 1 + \frac{3}{8} \cdot 10 + \frac{3}{8} \cdot 100 = \frac{83}{2}$

- $E[XY] = \frac{664}{8} = 83$

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 83 - 2 \cdot \frac{83}{2} = 0$$

$$(3) \quad P(X=2) = \frac{1}{4} \quad P(Y=1) = \frac{1}{4} \quad P(X=2, Y=1) = 0$$

$$P(X = 2, Y = 1) \neq P(X = 2)P(Y = 1)$$

$X \quad Y$

2023

p. 127

$$n$$

1

$$n$$

$$(n+1)$$

$$\underbrace{\binom{n}{1} \cdot \left(\frac{1}{6}\right)}_{2 \times 1} \cdot \underbrace{\left(\frac{1}{6}\right)^{n-1}}_{1 \times (n-1)} = \frac{n}{6^n}$$

4.**4.1.**

2025

2 $B(2, p)$

$$\begin{aligned}
 X &\sim B(2, p) & P(X = k) &= \binom{2}{k} p^k (1-p)^{2-k} \\
 E[X] &= \sum_{k=0}^2 k P(X = k) = P(X = 1) + 2P(X = 2) \\
 &= \binom{2}{1} p(1-p) + 2 \cdot \binom{2}{2} p^2 \\
 &= p[2(1-p) + 2p] = 2p \\
 \text{Var}(X) &= E[(X - E[X])^2] = E[(X - 2p)^2] \\
 &= \sum_{k=0}^2 (k - 2p)^2 P(X = k) \\
 &= (0 - 2p)^2 P(X = 0) + (1 - 2p)^2 P(X = 1) + (2 - 2p)^2 P(X = 2) \\
 &= 4p^2 \binom{2}{0} (1-p)^2 + (1 - 2p)^2 \binom{2}{1} p(1-p) + 4(1-p)^2 \binom{2}{2} p^2 \\
 &= 8p^2(1-p)^2 + 2p(1-p)(1-2p)^2 = 2p(1-p)
 \end{aligned}$$

4.2.

2024

p. 141

1 6

(1)	10	5
(2)	100	25

$$\begin{aligned}
 (1) \quad & X & X & \text{Bin}\left(10, \frac{1}{6}\right) \\
 P(X \geq 5) &= \sum_{k=5}^{10} P(X = k) = \sum_{k=5}^{10} \binom{10}{k} \left(\frac{1}{6}\right)^k \left(\frac{5}{6}\right)^{10-k} = \frac{1}{6^{10}} \sum_{k'=0}^5 \binom{10}{k'} 5^{k'} \\
 &= \frac{1}{6^{10}} (1 \cdot 1 + 10 \cdot 5 + 45 \cdot 25 + 120 \cdot 125 + 210 \cdot 625 + 252 \cdot 3125) \\
 &= \frac{934926}{60466176} = 0.01546... \approx 0.0155
 \end{aligned}$$

(2)

X

$n(1-p) > np = \frac{50}{3} > 5$

X

X

n

X

$Bin\left(100, \frac{1}{6}\right)$

$E[X] = \frac{50}{3}$

$\sigma[X] = \frac{5\sqrt{5}}{3}$

$-$

$N\left(100, \left(5\sqrt{5}/3\right)^2\right)$

$$P(X \geq 25) \approx P(X > 24.5) = P\left(Z > \frac{24.5 - 50/3}{5\sqrt{5}/3}\right) = P\left(Z > \frac{23.5}{5\sqrt{5}}\right)$$
$$= P\left(Z > 0.94\sqrt{5}\right) \approx P(Z > 2.102) \approx 1 - 0.9821 = 0.0179$$

4.3.

2023

2026

pp. 93 - 95, 141

p

X

2

$Bin(n, p)$

n

$q = \left(\frac{X}{n}\right)$

n

(1)

$P(|q - p| \leq 0.02) \geq 0.95$

(2)

(1)

n

$$\begin{aligned}
 P(|q - p| \leq 0.02) &= P\left(\left|\frac{q - p}{\sqrt{\text{Var}(q)}}\right| \leq \frac{0.02}{\sqrt{\text{Var}(q)}}\right) = P\left(|Z| \leq \frac{0.02}{\sqrt{\text{Var}(q)}}\right) \geq 0.95 \\
 \frac{0.02}{\sqrt{\text{Var}(q)}} &\geq 1.96 & P(|Z| \leq 1.96) &\approx 0.95 \\
 n &\geq \left(\frac{1.96}{0.02}\right)^2 p(1-p) = 9,604p(1-p) \geq 2,401
 \end{aligned}
 \tag{4}$$

$$\begin{aligned}
 &n \\
 &n \\
 &P(|q - p| \leq \varepsilon) \geq 1 - \alpha \\
 \sigma &= \sqrt{\frac{p(1-p)}{n}} \\
 P(|q - p| \leq \varepsilon) &= P\left(|q - p| < \frac{\varepsilon}{\sigma} \cdot \sigma\right) \geq 1 - \left(\frac{\sigma}{\varepsilon}\right)^2 \\
 P(|q - p| \leq \varepsilon) &\geq 1 - \left(\frac{\sigma}{\varepsilon}\right)^2 \geq 1 - \alpha \\
 \alpha &\geq \left(\frac{\sigma}{\varepsilon}\right)^2 = \frac{p(1-p)}{n\varepsilon^2} \Rightarrow n \geq \frac{p(1-p)}{\alpha\varepsilon^2} \geq \frac{1}{4\alpha\varepsilon^2} \\
 &- \\
 P(|q - p| \leq \varepsilon) &= P\left(\left|\frac{q - p}{\sigma}\right| \leq \frac{\varepsilon}{\sigma}\right) = P\left(|Z| \leq \frac{\varepsilon}{\sigma}\right) = 2\Phi\left(\frac{\varepsilon}{\sigma}\right) - 1 \\
 P(|q - p| \leq \varepsilon) &\geq 2\Phi\left(\frac{\varepsilon}{\sigma}\right) - 1 \geq 1 - \alpha \\
 \Phi\left(\varepsilon\sqrt{\frac{n}{p(1-p)}}\right) &\geq 1 - \frac{\alpha}{2} \\
 p & \quad z(p) \quad 1 - p = \Phi(z(p)) \\
 \varepsilon\sqrt{\frac{n}{p(1-p)}} &\geq z\left(\frac{\alpha}{2}\right) \Rightarrow n \geq \frac{p(1-p)}{\varepsilon^2} \left[z\left(\frac{\alpha}{2}\right)\right]^2 \geq \frac{1}{4\varepsilon^2} \left[z\left(\frac{\alpha}{2}\right)\right]^2
 \end{aligned}$$

4.4.

2024

pp. 143 - 145, 153

(1) :

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (-\infty < x < \infty)$$

$$y = f(x) \quad (x, y)$$

$$(2) \quad /$$

$$(3)$$

$$(1) \quad f' = -\frac{(x-\mu)}{\sigma^2} f \quad f'' = \frac{(x-\mu)-\sigma}{\sigma} \cdot \frac{(x-\mu)+\sigma}{\sigma} f$$

$$(2) \quad f(x)$$

$$(3)$$

$$0.5$$

4.5.

2022

pp. 121 - 122, 143

$$(1) \quad X \quad Y \quad N(170, 6^2)$$

$$(a) \quad W = \frac{X+Y}{2}$$

$$(b) \quad X \quad Y \quad W$$

$$(2) \quad X \quad Y$$

$$f(x) = \frac{1}{2\pi} \quad (0 \leq x \leq 2\pi), \quad g(x) = \frac{1}{12} \quad (-3 \leq x \leq 9)$$

$$W = XY$$

$$(1) \quad (a) \quad E[W] = E\left[\frac{1}{2}X + \frac{1}{2}Y\right] = \frac{1}{2}E[X] + \frac{1}{2}E[Y] = \frac{1}{2} \cdot 170 + \frac{1}{2} \cdot 170 = 170$$

$$(b) \quad \text{Var}(W) = \text{Var}\left(\frac{1}{2}X + \frac{1}{2}Y\right) = \left(\frac{1}{2}\right)^2 \text{Var}(X) + \left(\frac{1}{2}\right)^2 \text{Var}(Y) = \frac{1}{4} \cdot 6^2 + \frac{1}{4} \cdot 6^2 = 18$$

$$(2) \quad E[X] = \int_0^{2\pi} x f(x) dx = \frac{1}{2\pi} \int_0^{2\pi} x dx = \frac{1}{4\pi} [x^2]_0^{2\pi} = \frac{(2\pi)^2}{2\pi} = \pi$$

$$E[Y] = \int_{-3}^9 y g(y) dy = \frac{1}{12} \int_{-3}^9 y dy = \frac{1}{24} [y^2]_{-3}^9 = \frac{1}{24} (9^2 - (-3)^2) = 3$$

$$X \quad Y$$

$$E[W] = E[XY] = \pi \cdot 3 = 3\pi$$

4.6.

2024

pp. 147 - 151

(1) $X \sim N(\mu, \sigma^2)$

(a) $P(\mu - \sigma \leq X \leq \mu + \sigma)$

(b) $P(\mu - 2\sigma \leq X \leq \mu + 2\sigma)$

(c) $P(\mu - 3\sigma \leq X \leq \mu + 3\sigma)$

(2) $\Phi(z_0) = 0.01$ find z_0

(3)

(a) $z(0.01)$

(b) $\Phi(z_0) = 0.01$ find z_0

(c) $z(0.005)$

(1) $P(\mu + a\sigma \leq X \leq \mu + b\sigma) = \Phi(b) - \Phi(a)$ and $\Phi(a) - \Phi(-a) = 2\Phi(a) - 1$

(a) $P(\mu - \sigma \leq X \leq \mu + \sigma) = \Phi(1) - \Phi(-1) = 0.6826$

(b) $P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) = \Phi(2) - \Phi(-2) = 0.9544$

(c) $P(\mu - 3\sigma \leq X \leq \mu + 3\sigma) = \Phi(3) - \Phi(-3) = 0.9974$

(2) 3σ $[\mu - 3\sigma, \mu + 3\sigma]$ 3σ

(3) (a) 2.327

(b) -2.327

(c) 2.575

4.7.

2024

pp. 149 - 151

(1) $a \sim N(0, 1)$

$$P(|Z| \geq a) = 2P(Z \geq a)$$

(2) $X \sim N(m, \sigma^2)$

$$P\left(|X - m| \geq \frac{\sigma}{4}\right)$$
$$4$$
$$(3) \quad m \qquad \sigma \qquad n$$
$$\overline{X}$$
$$P\left(|\overline{X} - m| \geq \frac{\sigma}{4}\right) \leq 0.02$$
$$n$$

(1)

$f(x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}$

$f(-x) = f(x)$

$P(|Z| \geq a) = \int_{-\infty}^{-a} f(x)dx + \int_a^{\infty} f(x)dx = \int_a^{\infty} f(x)dx + \int_a^{\infty} f(x)dx = 2P(Z \geq a)$

(2)

$P\left(|X - m| \geq \frac{\sigma}{4}\right) = P\left(|Z| \geq \frac{1}{4}\right) \approx 0.8026 \approx 0.803$

(3)

$E[\overline{X}] = m \quad \sigma[\overline{X}] = \frac{\sigma}{\sqrt{n}}$

$P\left(|\overline{X} - m| \geq \frac{\sigma}{4}\right) = P\left(|Z| \geq \frac{\sqrt{n}}{4}\right) \leq 0.02$

$P(Z \leq 2.33) = 0.9901 \qquad P(|Z| > 2.33) \approx 0.0198 \approx 0.02$

$\frac{\sqrt{n}}{4} \geq 2.33 \implies n \geq (2.33 \cdot 4)^2 = 86.56$

$n = 87$

4.8.

2024

2026

pp. 149 - 150

4

E

350,198

66.31

23.55

X

$N(66.31, (23.55)^2)$

(1)

$P(X \geq 80)$

(2)

(3-a)

(3)

80

95

(4)

20%

26 of 48

(1)

$n = 350,198 \quad \mu = 66.31 \quad \sigma = 23.55$

X

$N(\mu, \sigma^2)$

$Z = \frac{X - \mu}{\sigma}$

$N(0, 1)$

$P(X \geq 80) = P\left(Z \geq \frac{80 - \mu}{\sigma}\right) \approx P(Z \geq 0.58)$

$P(X = 80) \approx 0$

$P(X \geq 80) = P(Z \geq 0.58) = 1 - \Phi(0.58)$

$\Phi(0.58) \approx 0.7910$

$P(X \geq 80) \approx 1 - 0.7910 = 0.2090$

(2)

E

1

80

E

80

(3)

X

$[80, 90]$

$P(80 \leq X \leq 90)$

$P(80 \leq X \leq 90) = P\left(Z \leq \frac{90 - \mu}{\sigma}\right) - P\left(Z \leq \frac{80 - \mu}{\sigma}\right)$

$= P\left(Z \leq \frac{90 - 66.31}{23.55}\right) - P\left(Z \leq \frac{80 - 66.31}{23.55}\right)$

$= \Phi(1.01) - \Phi(0.58) \approx 0.8438 - 0.7190 = 0.1248$

$nP(80 \leq X \leq 90) \approx 350,198 \times 0.1248 = 43,705$

(4)

20%

80%

φ

$P(X \leq \varphi) = 0.80$

$P(X \leq \varphi) = P\left(Z \leq \frac{\varphi - \mu}{\sigma}\right) = \Phi\left(\frac{\varphi - \mu}{\sigma}\right) \approx \Phi(0.84) \approx 0.7955 \doteq 0.80$

$\Phi\left(\frac{\varphi - \mu}{\sigma}\right) \approx \Phi(0.84)$

$\frac{\varphi - \mu}{\sigma} \approx 0.84$

$\varphi \approx$

$\mu + 0.84\sigma = 66.31 + 0.84 \times 23.55 = 86.0920 \approx 86$

86

4.9.

2025

pp. 149 - 150

4

120

10

100

15

C

(1)

95%

C

(2)

(1)

$X \sim N(120, 10^2)$

$) P(X \geq C) = \Phi\left(-\frac{C - 120}{10}\right)$

$$-\frac{C-120}{10} \approx 1.65 \Rightarrow C \approx 120 - 1.65 \times 10 = 103.5$$

$$(2) Y \sim N(100, 15^2) \quad P(Y \geq C) = 1 - \Phi\left(\frac{103.5 - 100}{15}\right) \approx 1 - 0.5910 = 0.4090$$

4.10.

2022

pp. 153 - 154

(1) 6

(2) 10000

(3) 10000

4900

5100

$$(1) \binom{6}{3} \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^3 = \frac{20}{2^6} = \frac{5}{16}$$

$$(2) \quad X \sim \text{Bin}\left(10000, \frac{1}{2}\right)$$

$$\begin{cases} E[X] = 10000 \times \frac{1}{2} = 5000 \\ \sigma[X] = \sqrt{10000 \times \frac{1}{2} \times \frac{1}{2}} = 50 \end{cases}$$

$$(3) np = n(1-p) = 5000 \quad - \quad X \quad N(5000, 50^2)$$

$$P(4900 \leq X \leq 5100) \approx P(4899.5 < X < 5100.5) = P(-2.01 <$$

$$Z < 2.01) \approx 0.9556$$

4.11.

pp. 93 - 95, 153 - 154

200

60

6

 X

(1)

$$42 \leq X \leq 78$$

(2)

(3)

 X

$$N(60, 6^2)$$

$$P(42 \leq X \leq 78)$$

$$n = 200 \quad \mu = 60 \quad \sigma = 6$$

(1)

$$P(42 \leq X \leq 78) = P(|X - 60| < 18) = P(|X - \mu| < 3\sigma) \geq 1 - \frac{1}{3^2} = \frac{8}{9}$$

$$(2) \quad Y = nX$$

$$Y = nP(42 \leq X \leq 78) \geq n \times \frac{8}{9} = 177.7...$$

178

$$(3) \quad X \sim N(\mu, \sigma^2)$$

$$P(42 \leq X \leq 78) = P(-3 \leq Z \leq 3) = 0.9974$$

5.

5.1.

2023

2025

pp. 168 - 169

2

$$Bin(n, p) \quad P(X = k) = \binom{n}{k} p^k (1 - p)^{n - k}$$
$$N(np, np(1 - p))$$
$$e^{-\lambda} \frac{\lambda^k}{k!}$$
$$\binom{n}{k} p^k (1 - p)^{n - k} \rightarrow$$

5.2.

2024

pp. 93, 168 - 169

1

ATM

ATM

1

40

ATM

1

60

ATM

/

60

75

90

105

120

$$M/M/1$$
$$\rho = \frac{\lambda}{\mu}$$
$$\rho < 1$$
$$L =$$
$$\frac{\rho}{1 - \rho}$$
$$W = \frac{1}{\lambda - \mu}$$
$$W = \frac{1}{1 - \frac{1}{90}} \text{ h} = \frac{1}{30} \text{ h} = 120 \text{ s}$$

5.3.

2023

pp. 166 - 169

$$(1) \quad 3 \quad 5$$

$$(2) \quad 1 \quad 2 \quad 1 \quad 4$$

$$(1) \quad 5 \quad X \quad X \sim \text{Bin}(5, 0.3)$$

$$P(X \geq 1) = 1 - P(X = 0) = 1 - (0.7)^5 = 1 - 0.16807 = 0.83193$$

$$(2) \quad 1 \quad X \quad X \sim \text{Po}(2)$$

$$P(X \geq 4) = 1 - P(0 \leq X \leq 3) = 1 - e^{-2} \left(1 + 2 + 2 + \frac{4}{3} \right) = 1 - \frac{19}{3} e^{-2}$$

$$\approx 1 - \frac{19}{3} \cdot \frac{18}{133} = \frac{57}{399} \approx 0.1429$$

$$e^2 = 1 + \frac{2}{(1-1) + \frac{1}{3 + \frac{1}{5 + \frac{1}{7 + \frac{1}{9 + \ddots}}}}} \approx 1 + \frac{2}{0 + \frac{1}{3 + \frac{1}{5 + \frac{1}{7 + 0}}}} = \frac{18}{399} \quad (5.1)$$

5.4.

2025

4

$$(1) \quad 1 \quad 3$$

2

$$(2) \quad X \quad P(X = 3) = 5P(X = 5)$$

$$P(X \leq 3)$$

$$(1) \quad X \sim \text{Po}(3)$$

$$P(X \geq 2) = 1 - P(0 \leq X \leq 1) = 1 - e^{-3}(1 + 3) = 1 - \frac{4}{e^3}$$

$$\approx 1 - 4 \cdot \frac{221}{4439} = \frac{3555}{4439} \approx 0.8009$$

$$e^3 \approx 1 + \frac{6}{(2-3) + \frac{9}{6 + \frac{9}{10 + \frac{9}{14 + \frac{9}{18+0}}}}} = \frac{4439}{221} \quad (5.2)$$

$$(2) \ P(X=3) = e^{-\lambda} \frac{\lambda^3}{3!} = e^{-\lambda} \frac{\lambda^3}{6} \quad 5P(X=5) = 5e^{-\lambda} \frac{\lambda^5}{5!} = e^{-\lambda} \frac{\lambda^3}{6} \left(\frac{\lambda}{2}\right)^2 \quad \lambda = 2$$

$$X \sim Po(2)$$

$$P(X \leq 3) = e^{-2} \left(1 + 2 + 2 + \frac{4}{3}\right) \approx \frac{18}{133} \cdot \frac{19}{3} = \frac{342}{399} \approx 0.8751$$

(5.1)

5.5.

2023

pp. 166 - 169

1%

200

(1) X

(2)

(3) 3

(2) (3)

$$(1) \ P(X=k) = \binom{n}{k} \left(\frac{1}{100}\right)^k \left(\frac{99}{100}\right)^{200-k} \approx e^{-2} \frac{2^k}{k!}$$

$$(2) \ P(X=0) \approx e^{-2} \approx \frac{18}{133} \approx 0.1353$$

$$(3) \ P(X \geq 3) = 1 - P(0 \leq X \leq 2) = 1 - e^{-2}(1 + 2 + 2) \approx 1 - 5 \cdot \frac{18}{133} = \frac{43}{133} \approx 0.3233$$

5.6.

2024

pp. 153 - 154, 166 - 169

4

(1) 240 6 38 45

(2) 0.1% 500 3

(1)

$n = 240$

$p = \frac{1}{6}$

6

X

$Bin(n, p)$

$np =$

40

$n(1 - p) = 200$

5

$-$

X

$N(np, np(1 - p)) = N\left(40, \frac{100}{3}\right)$

$P(38 \leq X \leq 45) \approx P(37.5 \leq X \leq 45.5) = P\left(\frac{37.5 - 40}{\frac{10}{\sqrt{3}}} \leq Z \leq \frac{45.5 - 40}{\frac{10}{\sqrt{3}}}\right)$

$= P\left(-0.25\sqrt{3} \leq Z \leq 0.55\sqrt{3}\right) \approx P(-0.43 \leq Z \leq 0.95)$

$= 0.8289 - 0.3336 = 0.4953$

(2)

$n = 500$

$p = \frac{1}{1000}$

X

$Bin(n, p)$

n

P

X

$Po(\lambda)$

$\lambda = np = \frac{1}{2}$

$P(X \geq 3) = 1 - P(0 \leq X \leq 2) = 1 - e^{-\frac{1}{2}}\left(1 + \frac{1}{2} + \frac{1}{8}\right) = 1 - e^{-\frac{1}{2}}\frac{13}{8}$

$\approx 1 - \frac{12 \cdot 2^2 - 6 \cdot 2 + 1}{12 \cdot 2^2 + 6 \cdot 2 + 1} \cdot \frac{13}{8} = 1 - \frac{37}{61} \cdot \frac{13}{8} = \frac{7}{488} \approx 0.0143$

$\sqrt[n]{e} \approx 1 + \frac{2}{(2n - 1) + \frac{1}{6n + 0}} = \frac{12n^2 + 6n + 1}{12n^2 - 6n + 1}$

(5.3)

5.7.

2026

$p = 0.001$

$n = 500$

X

- $e^{-0.5} \approx 0.6065,$

$\sqrt{0.5} \approx 0.7071$
- (1)

(a)

X

(b)

$E[X]$

$\text{Var}(X)$
- (2)

(a)

(b)

$P(X \geq 3)$

(3)

(a)

(b) $P(X \geq 3)$

(4)

(a) (2) (3)

(b)

np

(1) (a) $X \sim \text{Bin}(500, 0.001)$

(b) $E[X] = np = 0.5 \quad \text{Var}(X) = np(1-p) = 0.4995$

(2) (a) $n \quad p \quad X \quad \text{Po}(0.5)$

(b) $P(X \geq 3) = 1 - P(0 \leq X \leq 2) = 1 - e^{-0.5}(1 + 0.5 + 0.125) = 1 - e^{-0.5} \cdot 1.625 \approx 1 - 0.6065 \cdot 1.625 \approx 0.01444$

(3) (a) $E[X] = np = 0.5 \quad \text{Var}(X) = np(1-p) = 0.4995$

(b) $P(X \geq 3) \approx P(X \geq 2.5) = P\left(Z > \frac{2.5 - 0.5}{\sqrt{0.4995}}\right) \approx P\left(Z > \frac{2}{\sqrt{0.5}}\right) \approx P(Z > 2.83) \approx 0.0023$

(4) (a)

(b) $np = 0.5 \quad X$

5.8.

2022

2023

pp. 139 - 143, 168 - 169

$p \quad 0 < p < 1$

$(1-p)$

N

$N > 1$

(1) $N = 5$

(2) $t \quad 0 \leq t \leq N$

(3) m

$m =$

$N \quad 0 \leq m \leq N \quad m$

(4) $n \quad 0 \leq n \leq N \quad N - n \quad P(n \& N, p)$

$$(5) \quad (4) \quad n$$

$$(6) \quad n \quad \lambda = Np \quad N \rightarrow \infty$$

$$P(n; \lambda)$$

$$(4) \quad P(n \& N, p)$$

$$\lim_{N \rightarrow \infty} \left(1 + \frac{x}{N}\right)^N = e^x$$

$$(1) \quad p \cdot p \cdot (1-p) \cdot p \cdot (1-p) = p^3(1-p)^2$$

$$(2) \quad p^t$$

$$(3) \quad E[m] = \sum_{m=0}^{N-1} p^m(1-p) \cdot m + p^N \cdot m = \frac{p}{1-p}(1-p^N)$$

$$(4) \quad P(n \& N, p) = \binom{N}{n} p^n (1-p)^{N-n}$$

$$(5) \quad E[n] = \sum_{n=0}^N n P(n \& N, p) = Np$$

$$(6) \quad \lambda = Np$$

$$\lim_{N \rightarrow \infty} \binom{N}{n} p^n (1-p)^{N-n}$$

$$= \lim_{N \rightarrow \infty} \frac{N!}{n!(N-n)!} p^n (1-p)^{N-n}$$

$$= \lim_{N \rightarrow \infty} \frac{1}{n!} N(N-1)(N-2) \cdots (N-n+1) \left(\frac{\lambda}{N}\right)^n \left(1 - \frac{\lambda}{N}\right)^{N-n}$$

$$= \lim_{N \rightarrow \infty} \frac{\lambda^n}{n!} \left(1 - \frac{1}{N}\right) \left(1 - \frac{2}{N}\right) \cdots \left(1 - \frac{n-1}{N}\right) \left(1 - \frac{\lambda}{N}\right)^{-n} \left(1 + \frac{-\lambda}{N}\right)^N$$

$$= \frac{\lambda^n}{n!} e^{-\lambda}$$

6.

6.1.

2022

p. 174

2

(1)

(2)

(1)

(2)

6.2.

2025

(1) μ σ^2 n $X_1, X_2, \dots, X_n$.

(2)

(3) n $S_n = X_1 + X_2 + \dots + X_n$.

(4)

(1)

(2) $\forall \varepsilon > 0 : \lim_{n \rightarrow \infty} P(|\bar{X} - \mu| < \varepsilon) = 1$

(3) $\bar{X} = \frac{S_n}{n}$ n $N\left(\mu, \frac{\sigma^2}{n}\right)$

$S_n = n\bar{X}$ $N\left(n\mu, n^2 \frac{\sigma}{n}\right) = N(n\mu, n\sigma^2)$

(4)

6.3.

2025

p. 179

1180

20

(1) 25

1170

(2)

 N

0.9

1175

 N

(1) 25

$$\bar{X} \sim N\left(1180, \frac{20^2}{25}\right)$$

$$P(\bar{X} > 1170) = P\left(Z > \frac{1170 - 1180}{4}\right) = P(Z > -2.5) = 0.9938$$

(2)

 N

$$\bar{X} \sim N\left(1180, \frac{20^2}{N}\right)$$

$$P(\bar{X} > 1175) = P\left(Z > \frac{1175 - 1180}{20/\sqrt{N}}\right) = P\left(Z > -\frac{\sqrt{N}}{4}\right)$$

$$P(Z > -1.29) \approx 0.9015 > 0.9$$

$$-1.29 \geq -\frac{\sqrt{N}}{4}$$

$$\sqrt{N} \geq 1.29 \times 4 = 5.16 \quad N \geq (5.16)^2 = 26.6256$$

$$N \geq 27$$

6.4.

2025

1.2 kg

0.1 kg

100

122.5 kg

$$X_1, \dots, X_{100}$$

$$X = X_1 + \dots + X_{100}$$

$$E[X] = 100 \times 1.2 \text{ kg} = 120 \text{ kg}$$

$$\sigma[X] = \sqrt{100} \times 0.1 \text{ kg} = 1 \text{ kg}$$

$$X \sim N(120 \text{ kg}, 1 \text{ kg})$$

$$P(X < 122.5 \text{ kg}) = P\left(Z < \frac{122.5 - 120 \text{ kg}}{1 \text{ kg}}\right) = P(Z < 2.5) = 0.9938$$

6.5.

2022

pp. 202 - 205

μ σ^2 3 X_1 X_2 X_3

(1) .

(a) $T_1 = X_1 + X_2 - X_3$

(b) $T_2 = \frac{X_1 + 2X_2 + X_3}{4}$

(c) $T_3 = \frac{X_1 + X_2 + X_3}{3}$

μ

(2) $\{T_1, T_2, T_3\}$

(1) μ

(a) $E(T_1) = E(X_1) + E(X_2) - E(X_3) = \mu + \mu - \mu = \mu$

(b) $E(T_2) = \frac{1}{4}E(X_1) + \frac{2}{4}E(X_2) + \frac{1}{4}E(X_3) = \frac{1}{4}\mu + \frac{2}{4}\mu + \frac{1}{4}\mu = \mu$

(c) $E(T_3) = \frac{1}{3}E(X_1) + \frac{1}{3}E(X_2) + \frac{1}{3}E(X_3) = \frac{1}{3}\mu + \frac{1}{3}\mu + \frac{1}{3}\mu = \mu$

(2)

- (3)
- (4)

θ

$\Theta_n = T(X_1, X_2, \dots, X_n)$

3
- (a)

(b)

(c)

- (5)
- 3

- (1)

(a)

1.2

(b)

$\frac{1}{3}[(-1)^2 + 0^2 + 1^2] = \frac{2}{3}$

(c)

$\frac{3}{2} \times \frac{2}{3} = 1$
- (2)
- (3)
- (4)

(a)

$\hat{\theta}_n$

θ

$E[\hat{\theta}_n] = \theta$

(b)

n

$\hat{\theta}_n$

θ

$\forall \varepsilon > 0 : \lim_{n \rightarrow \infty} P(|\hat{\theta}_n - \theta| < \varepsilon) = 1$

(c)

$\hat{\theta}_n$

$\forall \hat{\theta}'_n : V(\hat{\theta}_n) \leq V(\hat{\theta}'_n)$

(5)

n

$n \rightarrow \infty$

6.7.

1000

10

32, 86, 78, 49, 48, 77, 97, 49, 51, 74

1000

$$\overline{X}$$
$$\overline{X} = \frac{1}{10}(32 + 86 + 78 + 49 + 48 + 77 + 97 + 49 + 51 + 74) = 64.1$$
$$\overline{X^2}$$
$$\overline{X^2} = \frac{1}{10}(32^2 + 86^2 + 78^2 + 49^2 + 48^2 + 77^2 + 97^2 + 49^2 + 51^2 + 74^2) = 4502.5$$
$$\sigma^2$$
$$\sigma^2 = \overline{X^2} - (\overline{X})^2 = 4502.5 - (64.1)^2 = 4502.5 - 4108.81 = 393.69$$
$$U^2$$
$$U^2 = \frac{10}{9}\sigma^2 = \frac{10}{9} \times 393.69 \approx 437.43$$

6.8.

2025

pp. 179, 216

p

n

X

(1) X

(2) n X

(3) p_0 95%

n

(4) $p = 0.05$ $n = 1900$ $P(76 \leq X \leq$

114)

(1) X $Bin(n,p)$

$$(2) \quad X \sim N(np, np(1-p))$$

$$(3) \quad 2z\left(\frac{\alpha}{2}\right)\sqrt{\frac{p_0(1-p_0)}{n}} = \frac{1}{2}z\left(\frac{\alpha}{2}\right)\sqrt{\frac{p_0(1-p_0)}{n}} \\ z\left(\frac{\alpha}{2}\right)\sqrt{\frac{p_0(1-p_0)}{n'}} = \frac{1}{2}z\left(\frac{\alpha}{2}\right)\sqrt{\frac{p_0(1-p_0)}{n}} \\ n' = \frac{1}{4}n$$

$$(4) \quad np = 1900 \cdot 0.05 = 95 \quad np(1-p) = 1900 \cdot 0.05 \cdot 0.95 = 90.25 = 9.5^2 \quad X \\ N(95, 9.5^2)$$

$$P(76 \leq X \leq 114) \approx P(75.5 < X < 114.5) = P\left(\frac{75.5 - 95}{9.5} < Z < \frac{114.5 - 95}{9.5}\right) \\ \approx P(-2.05 < Z < 2.05) \approx 0.9596$$

6.9.

2022

pp. 180 - 182

$$(1) \quad p = \frac{3500}{10000} = 0.35$$

$$(2) \quad p = 0.35$$

$$(1) \quad X \sim N(np, np(1-p)) \quad p = 0.35$$

$$\left[0.35 - 1.96\sqrt{\frac{0.35 \cdot 0.65}{10000}}, 0.35 + 1.96\sqrt{\frac{0.35 \cdot 0.65}{10000}}\right] \approx [0.3407, 0.3593]$$

$$p = 0.35$$

$$(2) \quad 10,000 \times 2^2 = 40,000$$

6.10.

2025

$$(1) \quad q = \frac{Y}{n} \quad A \quad p \quad 100(1-\alpha)\%$$

(2)

S

TV

T

140

32

T

- (a)
- p

95%
- (b)
- p

3%

0.95
- (c)

6.11.

2024

2024

pp. 223 - 225

(1)

$Bin(1, p)$

$X_1, X_2, ..., X_n$

- (a)
- (b)
- (2)
- n

A

Y
- (a)
- $q = \frac{Y}{n}$

A

p

$100(1 - \alpha)\%$
- (b)

(3)

S

TV

T

140

32

T

- (a)
- p

95%
- (b)
- p

3%

0.95
- (c)

6.12.

2024

pp. 223 - 225

(1)	p	0.8	0.9		0.99
	$\hat{p}$		p	0.02	
(2)	p			0.95	$\hat{p}$
	p		0.03		

6.13.

	2025				
2	p	S	$1 - p$	F	
m			S	k	$B(m; p)$
(1)	$B(m; p)$		$f(k)$	.	
(2)	X	$B(m; p)$	X	.	
(3)	p		$l(p)$	.	
(4)	p		.		
(5)		.	.		
	(a) (4)			?	
	(b) (4)			?	
	(c) (4)			?	

6.14.

	2025				
	$N(\mu, \sigma^2)$		μ	σ^2	
(1)	μ				
(2)		σ^2			

6.15.

	2025				
(1)					
(2)					
		/		σ^2	n
	N				
(3)		/			

7.

- ① 仮説の検定
- ② 統計量の検定
- ③ 優位水準と棄却域の設定
- ④ 帰無仮説の棄却・選択

7.1.

2022

2023

pp. 227 - 231

(1)

(2)

(3)

(4)

5%

(3)

5%

(5)

(1)

(1)

(6)

(6)

(7)

• (8:)

• (9:)

7.2.

2022

pp. 227 - 234

(1)

(2)

7.3.

2023	pp. 2235 - 236				
		8000			
		25		7700	
			5%		
		640			

7.4.

2023	pp. 238 - 240				
	T				
S		365			6
		330, 350, 385, 321, 340, 365			
	5%				

7.5.

2025	pp. 238 - 240				
	T				
S		360			6
		350, 355, 385, 390, 370, 365			

- (1)10%
- (2)5%

7.6.

2023	2024	2024	2025	pp. 235 - 236
(1)	μ		$N(\mu, 49)$	49
	$H_0 : \mu = 4,$	$H_1 : \mu > 4$		
	m			
	(a)	0.05	$m = 6$	2

(b)0.01 $m = 7$ 2

(2) $H_1 : \mu = 5$ 0.05

20.20.1

7.7.

2022pp. 241 - 242

(1) 2AB2

1012

A2.040.03B

2.020.022

5%

2

(2)AB

5%

7.8.

2022pp. 241 - 242

10070

6815

5%

7.9.

2024pp. 252 - 253

31

(1)A10A

“31”A

5%

(2) 107“31”

7.10.

2024pp. 252 - 253

5

1

5%

7.11.

2023

pp. 252 - 253

10

X

p

$H_0 : p = \frac{1}{2}$

H_0

(1) X

(2) X

(a) $H_1 : p < \frac{1}{2}$

5%

(b) $H_1 : p \neq \frac{1}{2}$

2%

7.12.

2023

pp. 252 - 253

5

1

p

(1) 5

5

0

(a) $p = \frac{1}{2}$

(b) $p = \frac{2}{3}$

(2) $p \geq \frac{1}{2}$

(a) 3

$p = \frac{1}{2}$

(b) $p = \frac{3}{4}$

“

3

”

7.13. 2

256

116

5%

(1) 2

(2) 2 10%

7.14.

2023 2025 pp. 252 - 253

$H_0 : p = \frac{1}{2}$

3

p

H_0

$0 < p < 1$

“ 3 3 ”

H_0

3

(1) 1 1

(2) 2 2

(3) 2 $\frac{1}{3}$ p

7.15.

2024 pp. 252 - 253

3

1

A B C

A 3 B 2 C 1

1

(1) A A $\frac{1}{16}$

B B $\frac{1}{9}$ C C $\frac{1}{25}$

1

(2) 960 640

(1) 5%